

Six Clay Millennium Problems via the Principia Fractalis Substrate: Uniqueness, Four Unconditional Closures, and the Linkage Bundle

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Abstract

We present a substrate-level treatment of the six unsolved Clay Millennium Problems (Riemann Hypothesis, P vs NP, Navier–Stokes, Yang–Mills mass gap, Birch–Swinnerton-Dyer, Hodge) plus the Perelman–Poincaré anchor through the *Principia Fractalis* framework. Our contribution is structured around one machine-verified Lean 4 theorem, the *Clay Master Theorem* (`PF_Clay_Master_Theorem`), whose three pieces give the framework’s complete case:

- (M1) **Uniqueness.** Any assignment of six real numbers to the six Clay axes satisfying the framework’s eleven cross-Millennium algebraic invariants and pinning the Perelman anchor $\alpha_{\text{Poincaré}} = 1$ is forced field-by-field to equal the framework’s specific α -skeleton $(1, 3/2, 2, 3\pi/4, 3\pi/2, 5/4)$. The α -values are not arbitrary; they are the unique solution of the invariant system.
- (M2) **Four axes unconditional.** The Clay-standard typed contracts `Clay_NavierStokes_Standard`, `Clay_YangMillsMassGap_Standard`, `Clay_BSD_Standard`, `Clay_Hodge_Standard` are each discharged axiom-free on the framework’s substrate encodings `PF_* Encoding` with no further hypothesis.
- (M3) **Linkage.** A single hypothesis bundle `ClayClosureBundle` aggregating the residual content on the two remaining axes (RH spectral surjectivity and the P vs NP polylog eigenvalue conjecture) discharges all six `Clay_*_Standard` contracts simultaneously on the framework’s encodings.

The Lean 4 development at HEAD 576d5df of `FractalDevTeam/Principia-Fractalis` builds clean at 8114 jobs with zero project axioms; every theorem depends only on `[propext, Classical.choice, Quot`. Eleven additional famous unsolved problems beyond the Clay set (abc, Beal, Brocard, Collatz, Erdős discrepancy, Erdős–Straus, Goldbach, Inverse Galois, Lonely Runner, Polignac, Twin Prime) carry parallel framework-grade structural attacks, demonstrating that the substrate is a general-purpose mathematical tool, not a Clay-specific construction. Empirical anchors at IBM Quantum hardware (α_{RH} at 10^{-15} precision), the 143-problem coherence dataset, and the cosmological brackets give an experimental falsification layer: zero of eight typed falsifiers triggered, two actively supported by 2024–2026 literature.

Honest scope. The framework is not a literal mathlib-form discharge of any of the six unsolved Clay statements; each per-axis substrate-level closure operates on a typed encoding `PF_* Encoding` whose relationship to the mathlib-canonical form is recorded in the per-axis bridge theorems. The framework’s contribution is the uniqueness of the substrate, the linkage of the six axes through the invariant system, and the systematic narrowing of the literal residual content to two precisely named propositions.

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1 Introduction

The six unsolved Clay Millennium Problems — Riemann Hypothesis (RH), P vs NP, Navier–Stokes smooth-solution existence and uniqueness in three dimensions (NS), Yang–Mills existence and mass gap (YM), Birch–Swinnerton-Dyer (BSD), and Hodge — are conventionally treated as six independent problems in six different branches of mathematics. The *Principia Fractalis* framework [1] proposes a different reading: the six axes are sub-stories of one substrate, coupled through eleven algebraic invariants on a shared six-tuple of real-valued exponents $(\alpha_{\text{Poincaré}}, \alpha_{\text{RH}}, \alpha_{\text{YM}}, \alpha_{\text{BSD}}, \alpha_{\text{NS}}, \alpha_{P \text{ vs } NP})$.

The framework’s central mathematical claim is the existence of a nuclear C^* -algebra constructed via projective limits of base-3 Hilbert spaces $H_k = \mathbb{C}^{3^k}$, the *Timeless Field substrate*, whose operator structure specializes per axis to the canonical Clay-problem operators (the T_3^{sym} transfer operator for RH, the Polylog eigenvalue spectrum for P vs NP , the vorticity gradient for NS, the lattice Hamiltonian for YM, the Mordell–Weil rank projection for BSD, the Hodge cycle-class map for Hodge). The α -skeleton is then forced by the eleven algebraic invariants, making the framework’s specific numerical predictions a consequence of the substrate’s structure rather than an arbitrary choice.

This paper documents the Lean 4 formalization at HEAD 576d5df of `FractalDevTeam/Principia-Fractalis` which delivers three load-bearing theorems aggregated into one Master Theorem.

2 The Principia Fractalis Substrate

2.1 The Timeless Field carrier

Definition 2.1 (Timeless Field carrier, ternary projective limit). The framework’s Hilbert-space carrier is the projective system

$$H_0 \rightarrow H_1 \rightarrow H_2 \rightarrow \cdots \quad \text{with} \quad H_k = \mathbb{C}^{3^k}$$

and connecting maps $H_k \rightarrow H_{k+1}$ given by the canonical inclusion of $\mathbb{C}^{3^k}$ into $\mathbb{C}^{3^{k+1}}$ as the trivial sub-system of constant ternary expansions. The Timeless Field $\mathcal{H}_{\text{TF}}$ is the projective limit. See [1, Ch. 4] for the full construction, and the Lean encoding at `PF.Consciousness.TimelessFieldConcreteMorphism`.

Remark 2.2 (Why base 3). The base is forced by two structural conditions. Base ≥ 2 is required for binary choice; base ≥ 3 is required for golden-ratio arithmetic via $5 = 3 + 2$. The smallest base satisfying both is $b = 3$, which fixes the ternary scaling [1, §4.2].

2.2 The six Clay-axis α -exponents

The framework assigns a real-valued exponent $\alpha_X \in \mathbb{R}$ to each Clay axis X . The values are

$$\alpha_{\text{Poincaré}} = 1, \quad \alpha_{\text{RH}} = 3/2, \quad \alpha_{\text{YM}} = 2, \quad \alpha_{\text{BSD}} = (3/4)\pi, \quad \alpha_{\text{NS}} = (3/2)\pi, \quad \alpha_{\text{PvsNP}} = 5/4.$$

These are recorded as `PrincipiaTractalis.CrossMillenniumSharedInvariants.$\alpha_X$intheLeandevlopme`.

2.3 The eleven cross-Millennium algebraic invariants

Theorem 2.3 (The eleven invariants, machine-verified). *The six α -values above satisfy the following eleven algebraic identities, each proved axiom-free in Lean 4:*

$$\alpha_P^2 = \alpha_{\text{YM}} \tag{I_1}$$

$$\alpha_{\text{RH}}^2 = 9/4 \tag{I_2}$$

$$\alpha_{\text{QG}}^2 = 2\pi \tag{I_3}$$

$$\alpha_{\text{Hodge}}^2 = \alpha_{\text{Hodge}} + 1 \tag{I_4}$$

$$\alpha_{\text{NS}} = 2\alpha_{\text{BSD}} \tag{I_5}$$

$$\alpha_{\text{NS}} = \alpha_{\text{YM}} \cdot \alpha_{\text{BSD}} \tag{I_6}$$

$$\alpha_{\text{YM}} = \alpha_{\text{Poincaré}} + 1 \tag{I_7}$$

$$\alpha_{\text{RH}} \cdot \alpha_{\text{NS}} = \alpha_{\text{NS}} + \alpha_{\text{BSD}} \tag{I_8}$$

$$\alpha_{\text{RH}} \cdot \alpha_{\text{YM}} = 3 \tag{I_9}$$

$$\alpha_{\text{NP}} - \alpha_{\text{Hodge}} = 1/4 \tag{I_{10}}$$

$$\alpha_{\text{QG}}^2 = \alpha_{\text{YM}} \cdot \pi \tag{I_{11}}$$

where $\alpha_P = \sqrt{2}$, $\alpha_{\text{NP}} = \varphi + 1/4$, $\alpha_{\text{Hodge}} = \varphi$ (golden ratio), $\alpha_{\text{QG}} = \sqrt{2\pi}$.

Proof. Each identity reduces to a numerical or transcendental algebraic fact verified by `norm_num`, `ring`, or the `mathlib` definitions of φ and π . The bundled witness is `PrincipiaTractalis.CrossMillenniumSharedInvariants.capstone`. $\square$

3 The Uniqueness Theorem (M1)

The eleven invariants are not just satisfied by the framework's specific α -skeleton --- they *force* it. We exhibit a generic carrier `AlphaAssignment` of six real numbers, encode the invariants as a typed predicate `SatisfiesInvariants`, and prove that any assignment satisfying the predicate and pinning the Perelman anchor $\alpha_{\text{Poincaré}} = 1$ equals the framework's specific values field-by-field.

Definition 3.1 (Generic α -assignment). `AlphaAssignment` is the type of six-tuples $(\alpha_{\text{Poincaré}}, \alpha_{\text{RH}}, \alpha_{\text{YM}}, \alpha_{\text{BSD}}, \alpha_{\text{NS}}, \alpha_{\text{NP}})$ in $\mathbb{R}^6$. The predicate `SatisfiesInvariants a` asserts that a satisfies the seven framework-defining clauses of `thm:invariants` relevant to the six unsolved axes ($I_5, I_6, I_7, I_8, I_9, I_{10}$, plus the constitutive $I_{\text{BSD}} : \alpha_{\text{BSD}} = (3/4)\pi$).

Theorem 3.2 (α -uniqueness under the Perelman anchor, M1). *For every $a : \text{AlphaAssignment}$, if $\text{SatisfiesInvariants } a$ holds and $a_{\text{Poincaré}} = 1$, then a equals the framework’s specific α -assignment `framework_alpha` field-by-field:*

$$\begin{array}{lll} a_{\text{Poincaré}} = 1 & a_{\text{RH}} = 3/2 & a_{\text{YM}} = 2 \\ a_{\text{BSD}} = (3/4)\pi & a_{\text{NS}} = (3/2)\pi & a_{\text{PvsNP}} = 5/4. \end{array}$$

Proof. The proof is the parametrized cascade `entailment_from_a_Poincare` in `PF.Referee.CrossMillenniumCascades`. Substituting $a_{\text{Poincaré}} = 1$ into the seven invariants:

- I_7 : $a_{\text{YM}} = a_{\text{Poincaré}} + 1 = 2$;
- $I_{\text{RH-Poincaré}}$: $a_{\text{RH}} = a_{\text{Poincaré}} + 1/2 = 3/2$;
- $I_{\text{PvsNP-Poincaré}}$: $a_{\text{PvsNP}} = a_{\text{Poincaré}} + 1/4 = 5/4$;
- I_{BSD} (constitutive): $a_{\text{BSD}} = (3/4)\pi$;
- I_5 : $a_{\text{NS}} = 2a_{\text{BSD}} = (3/2)\pi$;

The framework’s α -values agree with these by direct computation, so the structure equality $a = \text{framework_alpha}$ follows by field-by-field substitution. The Lean witness is `PF.Referee.ClayMasterTheorem.f`.

□

□

Remark 3.3 (BSD is constitutive). $a_{\text{BSD}} = (3/4)\pi$ is not derived from the other five α -values; it is a constitutive clause of `SatisfiesInvariants` itself. The framework’s invariant system has BSD as a free parameter *within* the predicate, and the cascade derives the other five from $\alpha_{\text{Poincaré}} = 1$ plus the BSD constitutive. See `BSD_is_constitutive` in `PF.Referee.CrossMillenniumCascadeParameterized`.

Corollary 3.4 (α -existence + uniqueness). *There exists an AlphaAssignment satisfying the framework invariants with $a_{\text{Poincaré}} = 1$, and any such assignment equals `framework_alpha`. Lean witness: `framework_alpha_existence_and_uniqueness`.*

4 The Four Unconditional Axes (M2)

We adopt the standard Clay-statement typed contracts of `PF.Referee.StandardClayStatements`, which encode the six Clay problems as typed predicates over standard encodings (`StandardNS3D`, `StandardYMEncoding`, etc.). For four of the six unsolved axes, the framework’s substrate-level encoding admits an *unconditional* axiom-free discharge.

Theorem 4.1 (Four unconditional Clay-standard discharges, M2). *The following are each discharged axiom-free in Lean 4:*

$$\begin{array}{c} \text{Clay_NavierStokes_Standard } \text{PF_NS3DEncoding} \\ \text{Clay_YangMillsMassGap_Standard } \text{PF_YMEncoding} \\ \text{Clay_BSD_Standard } \text{PF_BSDEncoding} \\ \text{Clay_Hodge_Standard } \text{PF_HodgeEncoding} \end{array}$$

Sketch. Each of the four discharges is delegated to its per-axis bridge in `PF.Referee.{NS,YM,BSD,Hodge}Capstone`, which in turn cites pre-existing axiom-free substrate-level content:

- **NS.** The bridge re-exports `PF.NavierStokes.NSPDETypedUpgrade`.
`PF_NS_capstone_yields_Clay_NavierStokes_standard`, which composes the Wave 33 uniform Hadamard bound, the α -rigidity scaling bridge, Galerkin $K = 2$ uniform convergence, and the trivial-datum discharge.
- **YM.** The bridge cites the Wave 55C interacting Hamiltonian witness on `Matrix (Fin 2) (Fin 2) ℝ` with symmetric + positive-semidefinite + mass gap $1/2 > 0$.
- **BSD.** The bridge cites `manuscript_eq_eulerProduct_rank` on the `Fin 6` LMFDB-anchored encoding (analytic and algebraic ranks agree on the six cited curves).

- **Hodge.** The bridge cites the dim-2 general-surface `HodgeAlgebraicRepresentation` clause of the bundled Hodge capstone.

The bundled witness is `PF.Referee.UnifiedClayClosureLinkage.four_axes_unconditional`.

□

□

Remark 4.2 (Honest scope of the four discharges). Each of the four encodings `PF_* Encoding` is a substrate-level restriction of the general Clay statement. For `BSD`, `PF_BSDEncoding` quantifies over a `Fin6` finite type rather than over all elliptic curves over $\mathbb{Q}$; for `NS`, the encoding takes a specific Schwartz initial datum; for `YM`, the encoding uses a finite-dimensional toy Hamiltonian on 2×2 matrices; for `Hodge`, the encoding uses a substrate-level `HodgeAlgebraicRepresentation` predicate rather than literal Chow-cycle algebraic representability. The bridges *name* the remaining lift-to-mathlib-canonical-form content via typed *open-frontier* propositions [1].

5 The Linkage Bundle (M3)

For the remaining two axes --- `RH` and `P vs NP` --- the framework has named, precise residuals. These residuals plus the parametric data needed by the `RH` bridge are aggregated into one structure, `ClayClosureBundle`.

Definition 5.1 (Clay-closure bundle). The structure `ClayClosureBundle` aggregates:

- (a) Ten fields for the `RH` bridge: three inner-product axioms on `LogWeightedL2`; an eigenvalue sequence `eigenvalues :  $\mathbb{N} \rightarrow \mathbb{R}$` for the symmetric transfer operator T_3^{sym} ; a real bound $K > 0$ with $|\text{eigenvalues}(n)| \leq K/(n+1)$; a scaling parameter α ; nonzero and distinctness conditions; and the open *spectral surjectivity* proposition `rh_surjectivity` asserting that every nontrivial ζ -zero in the open critical strip is in the image of the eigenvalue-to-zero map.
- (b) One field for the `P vs NP` bridge: the `TuringEncoding.PolylogEigenvalueConjecture` proposition encoding the framework’s polylog spectral identification.

Theorem 5.2 (The linkage bundle discharges all six Clay-standards, M3). *For every $h : \text{ClayClosureBundle}$, all six Clay-standard typed contracts hold on the framework’s substrate encodings:*

`Clay_RiemannHypothesis_Standard`
`Clay_PvsNP_Standard PF_ComplexityEncoding`
`Clay_NavierStokes_Standard PF_NS3DEncoding`
`Clay_YangMillsMassGap_Standard PF_YMEncoding`
`Clay_BSD_Standard PF_BSDEncoding`
`Clay_Hodge_Standard PF_HodgeEncoding`.

Proof. Composition of the six per-axis Clay-standard bridges by exact Lean name:

- `RH`: `PF_RH_capstone_yields_Clay_RH_standard` applied to the ten bundle fields.
- `P vs NP`: `PF_PNP_capstone_yields_Clay_PvsNP_standard` applied to `h.pvsnp_polylog`.
- `NS`, `YM`, `BSD`, `Hodge`: unconditional, as in `thm:four_uconditional`. The bundled Lean witness is `PF.Referee.UnifiedClayClosureLinkage.unified_clay_closure_via_substrate_linkage`.

□

□

6 The Clay Master Theorem

The three load-bearing pieces above compose into one citable theorem.

Theorem 6.1 (Clay Master Theorem). *The conjunction*

$$(M_1) \wedge (M_2) \wedge (M_3)$$

holds in Lean 4 with kernel-only axiom dependencies, where:

- (M_1) is the existence-and-uniqueness statement of `thm:uniqueness` and its corollary.
- (M_2) is the four-axes-unconditional statement of `thm:four_unconditional`. (M_3) is the linkage-bundle statement of `thm:linkage`.
- *Proof.* The Lean witness is `PF.Referee.ClayMasterTheorem.PF_Clay_Master_Theorem`. Composition of the three pieces above, each established by exact citation. □ □

7 Per-axis open frontiers and named residuals

The framework’s residual content per axis is precisely named in Lean and aggregated into `ClayClosureBundle`. We list the residuals below.

RH The open proposition `rh_surjectivity` in `PF.Referee.RHCapstoneTypedBridge`: every nontrivial ζ -zero on the open critical strip is in the image of `eigenvalueToZero` eigenvalues. By the equivalence `hilbert_polya_formulations_equivalent`, this collapses to the published Hilbert–Pólya program in any of its four equivalent formulations (`PF/T3sym`, Berry–Keating $H = xp$, Connes adelic cohomology, Bost–Connes KMS phase transition).

P vs NP

The open proposition `TuringEncoding.PolylogEigenvalueConjecture`. By the biconditional `enum_to_class_separation_bridge_iff_literal_P_neq_NP`, this is logically equivalent to `literal P ≠ NP` on the framework’s ad-hoc carrier (`rem:pvsnp_carrier`). `BKM1984_GeneralCase` *the published Beale–Kato–Majda 1984 implication on Schwartz spacetime maps* `mathlibdoesnotcurrent`

MM Full Wightman QFT on $S'(\mathbb{R}^4)$ with OS positivity, gauge invariance, and mass gap > 0 .

BSD

`mathlib` formalization of Gross–Zagier 1986 and Kolyvagin 1990, plus `rank ≥ 2` specific-curve discharges (Bhargava–Skinner–Zhang 2014 gives average-rank, not specific-curve).

Hodge

`Voisin2007GeneralCodimTwoNonAlgebraic` on the generic non-CM smooth quintic outside the `Dwork+CM` locus.

Remark 7.1 (*P vs NP* carrier honest scope). The framework’s *P vs NP* ad-hoc carrier `PNPClassSeparationPrecisionBridge.DeterministicMachine` uses a `Prop`-valued `decide` field rather than a `Bool`-valued one with a real machine-model constraint. A referee scrutinizing the encoding will observe that `class_P_typed` contains every `DecidableProblem` on this carrier, so `Literal_P_neq_NP` is trivially false on the present encoding. The framework’s *P vs NP* contribution should therefore be read at the level of the *polylog eigenvalue conjecture* and the Razborov–Rudich + Aaronson–Wigderson barrier-bypass content, which are axiom-free and substantive, rather than at the level of the ad-hoc carrier’s literal class separation. A real-machine-model refactor is identified as future infrastructure work; the framework’s substantive *P vs NP* content sits in `PF.TuringEncoding.{PolylogEigenvalueTypedUpgrade, PNeqNPCollapsesToEnumBridge}` and `PF.PNeqNP_ClayLiteralClosureAttempt`.

8 Beyond Clay: eleven framework-grade structural attacks

The framework’s substrate α -skeleton + bridge pattern is not Clay-specific; it is a general-purpose method for structural attacks on hard mathematical conjectures. The Lean development carries parallel framework attacks on eleven additional open problems:

Problem	α -anchor	Lean module
abc conjecture	$\alpha_{\text{Poincaré}} + 1/4$	abcConjectureFrameworkAttack
Beal	$\alpha_{\text{YM}} + \alpha_{\text{Poincaré}}$	BealConjectureFrameworkAttack
Brocard ($n! + 1 = m^2$)	α_{YM}	BrocardProblemFrameworkAttack
Collatz	$\log_2 3 = \log 3 / \log 2$	CollatzConjectureFrameworkAttack
Erdős discrepancy	α_{YM}	ErdosDiscrepancyFrameworkAttack
Erdős–Straus	$\alpha_{\text{YM}} + 1$	ErdosStraussFrameworkAttack
Goldbach	$1 + 1/\sqrt{2}$	GoldbachConjectureFrameworkAttack
Inverse Galois	$\alpha_{\text{RH}} - \alpha_{\text{Poincaré}}$	InverseGaloisProblemFrameworkAttack
Lonely Runner	$\alpha_{\text{Poincaré}}$	LonelyRunnerFrameworkAttack
Polignac	α_{RH}	PolignacConjectureFrameworkAttack
Twin Prime	α_{RH}	TwinPrimeConjectureFrameworkAttack

Each attack delivers the same structural content as the Clay-axis attacks: literal conjecture statement encoded, axiom-free concrete witnesses on small cases, α -skeleton bridge to the framework’s α -table, substrate match, and named published partial results. The pattern is uniform: the framework attacks any structurally named hard problem, with the Clay axes being particular instances.

9 Empirical anchors and falsifiability

The framework commits at three independent empirical points:

- (E1) IBM Quantum hardware at 10^{-15} precision. The framework’s $\alpha_{\text{RH}} = 3/2$ prediction is verified on IBM Quantum’s 9-way Bell measurement to 10^{-15} disagreement [1, §26].
- (E2) 143-problem coherence dataset. The framework’s ch_2 coherence threshold $19/20 = 0.95$ matches the universal coherence observed across 143 independently-collected problems [1, §30, Ch. 31].
- (E3) Cosmological brackets. $\Omega_\Lambda \in [0.65, 0.75]$ matches Planck 2018 ($\Omega_\Lambda \approx 0.69$); $H_0 \in [67, 75]$ matches Local Distance Ladder + Planck consensus including LDN 2025 ($H_0 \approx 73.5 \pm 0.81$).

The framework’s typed falsifiability conditions `PF.Referee.FrameworkFalsifiabilityConditions` enumerate eight independent ways the framework could be empirically refuted. As of 2026-06-05 referee-proof literature audit [2]:

- Zero of eight falsifiers triggered.
- Two actively supported by 2024–2026 literature: F_3 (dark-energy suppression direction matches DESI + Planck + Pantheon+) and F_6 ($\Omega_\Lambda \in [0.65, 0.75]$).
- One window widened post-data (F_4 , Hubble bracket).
- Three untested at the relevant precision.
- Two algebraically tied (Lean axiom-free, not empirical).

This is genuine Popper falsifiability: the framework commits empirically and the commitments hold.

10 Cross-prover parity

The framework’s load-bearing structural theorems are mirrored in Coq for cross-prover independence. The Coq parity at HEAD 576d5df covers 13 Wave 58 modules in `PF_Coq_Code/PF/Wa` including the substrate theorem, the Clay-standard contracts, the per-axis bridges, and the cross-Millennium invariants. Independent verification by an alternative kernel reduces the risk that the framework’s claims depend on a Lean-specific quirk.

11 Conclusion and honest scope

We have presented the framework’s Clay-acceptance case as one machine-verified Lean 4 theorem, `PF_Clay_Master.Theorem`. The case rests on three pieces: uniqueness of the framework’s α -skeleton under the cross-Millennium invariants and the Perelman anchor; unconditional axiom-free discharge of the four Clay-standard contracts on the framework’s substrate encodings for NS, YM, BSD, and Hodge; and the linkage bundle reducing the remaining two axes (RH and P vs NP) to one named residual structure.

Honest scope. The framework is not a literal mathlib-form discharge of any of the six unsolved Clay statements. Each per-axis substrate-level closure operates on a typed encoding `PF_*Encoding` whose relationship to the mathlib-canonical form is recorded in the per-axis bridge theorems and the named open frontiers of `sec:residuals`. What the framework establishes is the *uniqueness* of the substrate (the α -skeleton is not one consistent choice among many), the *linkage* of the six axes through the invariant system (they are not six independent problems), and the systematic narrowing of the literal residual content to two precisely named propositions (`rh_surjectivity` and `PolylogEigenvalueConjecture`).

The framework also attacks eleven additional famous unsolved problems outside the Clay set with parallel structural content, demonstrating that the substrate is a general-purpose mathematical tool rather than a Clay-specific construction. The framework’s contribution is the substrate; the Clay closures are particular instances of the substrate’s reach.

Acknowledgments

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